Assignment

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Class: BS Computer Science

Section: E

Roll No: 12

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Subject: PF

Lab No: 05

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Lab 05

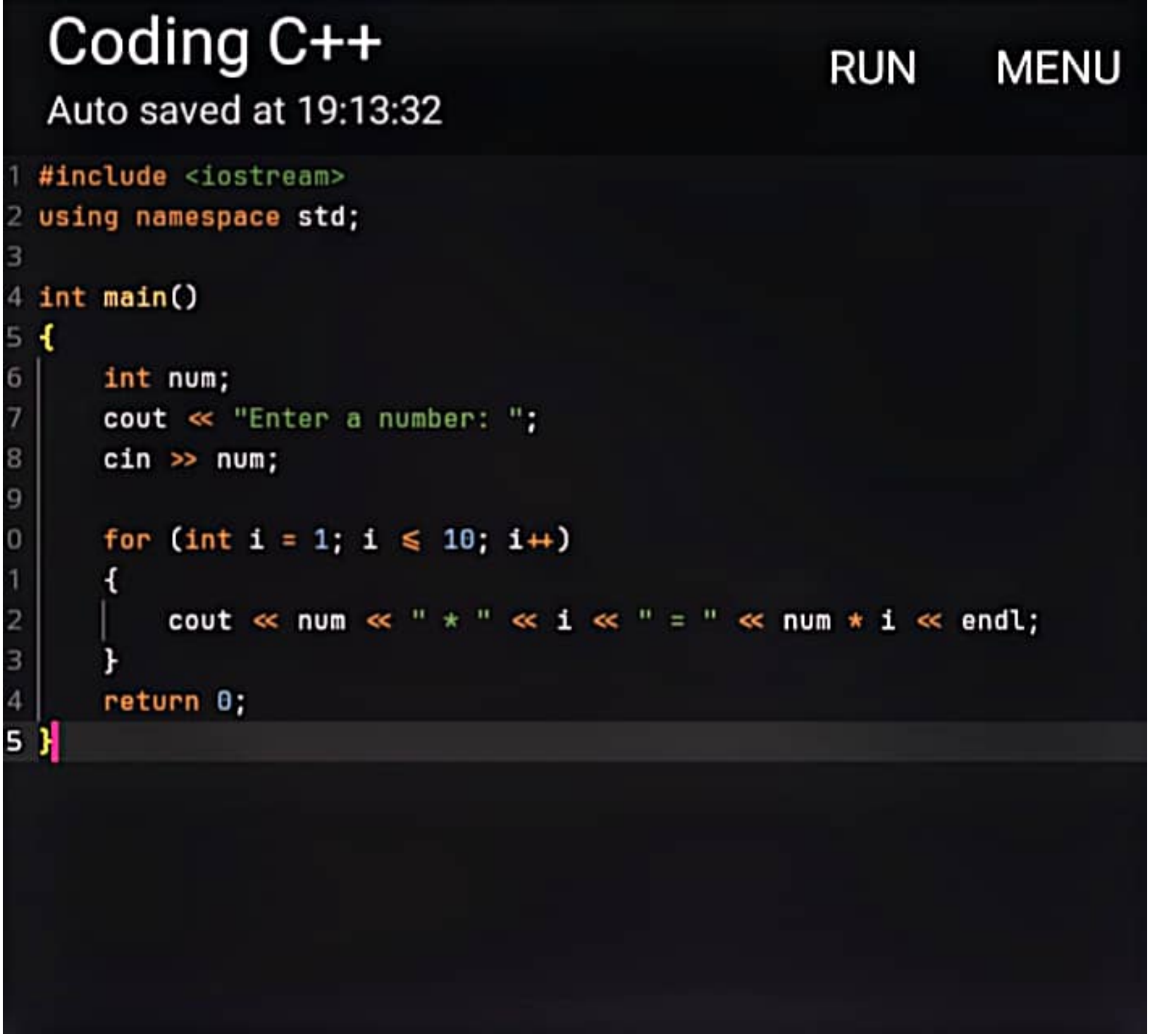
Task 1

```
# include <iostream>
  Using namespace std;
     int main ( )  {
int num;
 cout << "Enter a number: ";
 Cin >> num;
 Couti For (int ; = 1; 1 <=10; i + +){
Cout << num << " x " << "; << = + num i end}
return 0;
     }
```

Output..

2 * 1 = 2
2 * 2 = 4
2 * 3 = 6
2 * 4 = 8
2 * 5 = 10
2 * 6 = 12
2 * 7 = 14
2 * 8 = 16
2 * 9 = 18
2 * 10 = 20

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```
Enter a number: 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

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Task#2

```
#include <iostream>
using namespace std;
int main ()  {
int num, power result = 1;
cout << "Enter a number;
cin >> num;
cout << "Enter" power";;
cin >> power,
for cin; = 1; ;< = power; i++) {result= result * num
cout << "Answer: << result;
```

Output:

Enter number= 2 :: Answer=8

Enter power = 3

Task # 3

```
# include <iostream>
using namespace std;
int main ()  {
    int n, sum = 0;
cout << "Enter a number";;
cin >> n;
for (int i=1;;<=n i++) {sum+=i;}
cout <<" sum =" <<Sum;
return 0;}
```

Output:-

Enter a number: 5

Sum : 15

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```
Enter base: 2
Enter power: 3
2 ^ 3 = 8
```

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```
Enter a number: 10
Sum of first 10 natural numbers = 55
```

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Task #4

```
#include <iostream>
Using namespace std;
int main () {
    int num, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    while (num > 0) {
    Sum = sum + num % 10;
    num = num / 10
    cout << "Sum of digit = " << sum;
    return 0; }
```

Output:-

Enter a number: 358

Sum of digit : 16

Task #5

```
#include <iostream>
Using namespace std;
int main () {
    int num, fact = 1;
    Cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i <= num; i++) {
    fact = fact * 1; }
    cout << "factorial" = << fact;
    return 0; }
```

Enter a number : 4

factorial : 24

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Coding C++

Auto saved at 19:16:07

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num, sum = 0, remainder;

    cout << "Enter a number: ";
    cin >> num;

    for (; num > 0; num = num / 10)
    {
        remainder = num % 10;
        sum = sum + remainder;
    }

    cout << "Sum of digits = " << sum;
    return 0;
}
```

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```
Enter a number: 358
Sum of digits = 16
```

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Coding C++

Auto saved at 19:16:58

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;
    long fact = 1;

    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= num; i++)
    {
        fact = fact * i;
    }

    cout << "Factorial of " << num << " = " << fact;
    return 0;
}
```

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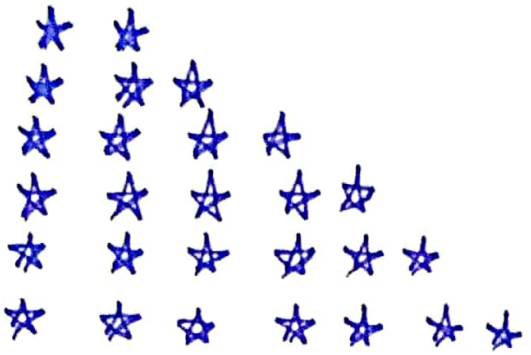
```
Enter a number: 4
Factorial of 4 = 24
```

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Task # 6

```
# include <iostream>
Using namespace std;
 int main () {
 for(int i=; 1; i<=7; 1++){
  for Int j= 1; j<=i; J++) {
 Cout << "*";; }
 Cout<< end1; }
  return 0:  }
```

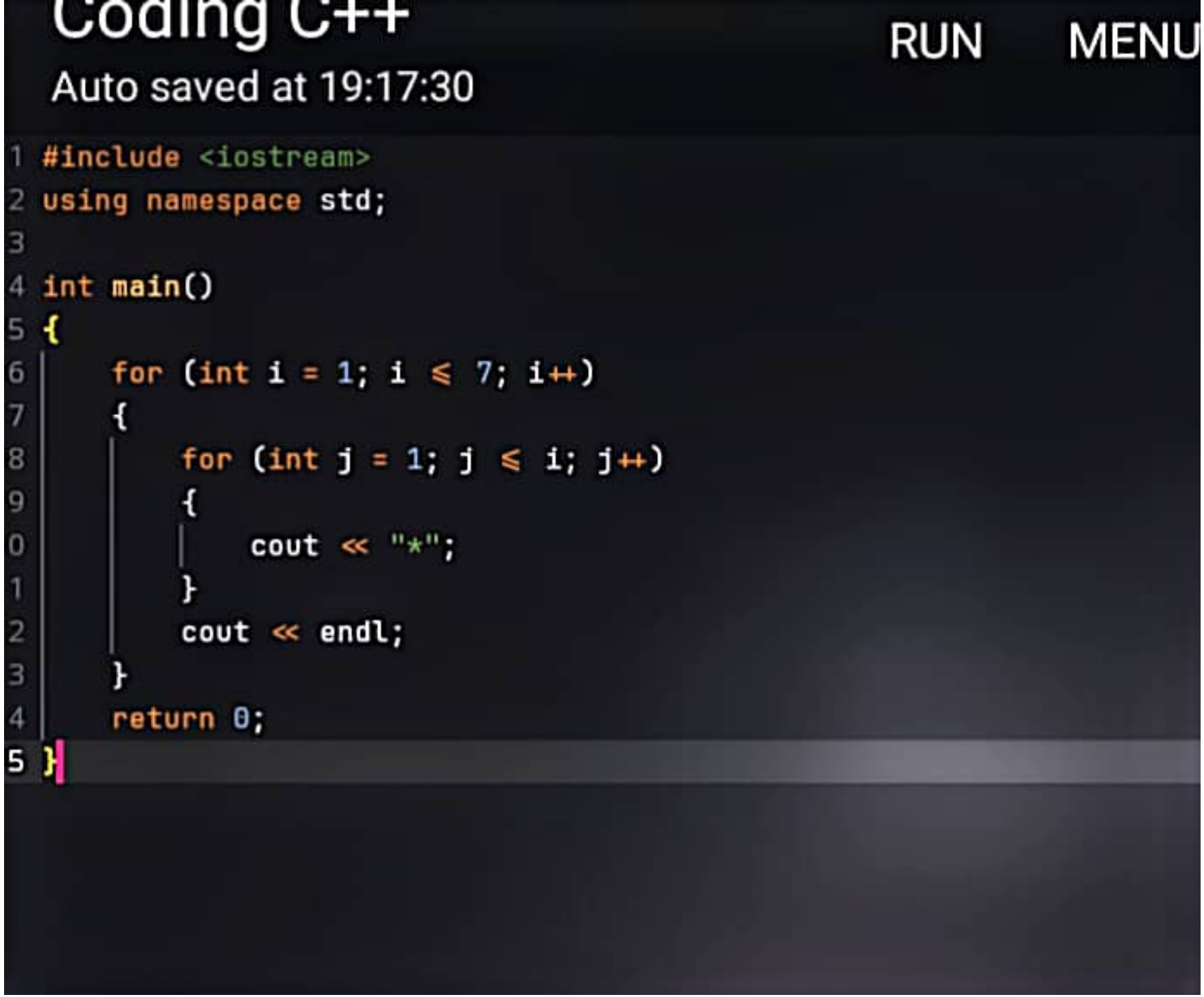
Output: *



Task # 7

```
# include <iostream>
Using namespace std;
 int main () {
    int month;
   Cout << "Enter month number";;
   Cin >> month
    Switch (month) {
 Case 1; Case 3; Case 5; Case 7; case 8; Case 10; Case 12;
```

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```
cout<< "31 days";
   break;
Case 4; Case 6; case 9; Case 11;
 Cout <<" 30days": ;
 case 2;
 cout <<"25 days or 29days";
  break
  default;
  cout<< "invalid month": ;}
   return 0;

    }
```

output:

Enter month number: 2

25 or 29 days

Task # 8

```
# include <iostream>
 Using namespace std;
  int main () {
  char ch;
  cout <<"Enter an alphabet:";
  Cin >> ch;
  Cout <<" swith ch {
 case a' case e; Case i; case o; case u;
```

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Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int month;
    cout << "Enter month number (1-12): ";
    cin >> month;

    switch (month)
    {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            cout << "31 days";
            break;
        case 4:
        case 6:
        case 9:
        case 11:
            cout << "30 days";
            break;
        case 2:
            cout << "28 days";
            break;
        default:
            cout << "Invalid month";
    }
    return 0;
}
```

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```
Enter month number (1-12): 2
28 days
```

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```
case 'A': Case; case 'I'; case 'O': case 'U';
cout << "Vowel";;
 default;
 if ((ch>='B' && ch<='Z') || (ch>='a' && ch<='z')
 cout << "consonant";
 else
 cout << "Invalid input;}
 return 0;
        }
```

Output:.

Enter an alphabet : A

Vowel

Think Section

→ code1 output: Hello Hello Hello.....

Explanation The condition 2 is always trul so it contirte printing Hello indefinety until the program is stoppeh

→ Code 2 output: No output

Explanation: value of i is 0. The condition is i<9 is false so loop never excetes.

→ Code 3 output: 1 0 1 0 1 0 1 0 1 0 1 0.....

Explanation The for (;;) loop has no initillit ration condition or increment.

→ Code 4 out: c++ for loop 12345

Explanation:- first the program print c++ for loop that the loop print the number to 5 because the condition is x<=fun and fun =5

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Coding C++

RUN MENU

Auto saved at 19:18:37

```cpp
#include <iostream>
using namespace std;

int main()
{
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;

    switch (ch)
    {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
            cout << ch << " is a Vowel";
            break;
        default:
            cout << ch << " is a Consonant";
    }
    return 0;
}
```

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```
Enter an alphabet: e
e is a Vowel
```

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